Chapter 3
Alkanes and Cycloalkanes: Conformations and cis-trans Stereoisomers

Additional Recommended Problems: Chp 3 – 23-31, 34-37, 42, 43

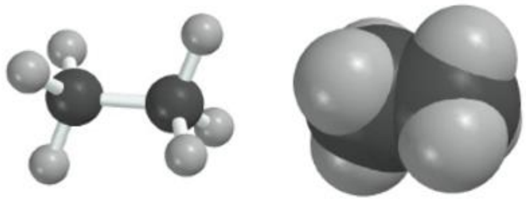
Conformation of Ethane

- Ethane (H_3CCH_3) an infinite number of conformations related by rotation about the C–C bond
- The two most important are the **staggered** and **eclipsed** conformations

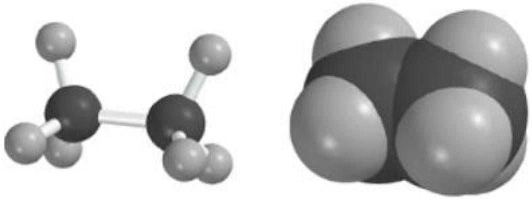
- **Staggered conformation:** each C–H bond *bisects* an H–C–H

- **Eclipsed conformation:** each C–H bond is *aligned* with an adjacent C–H bond

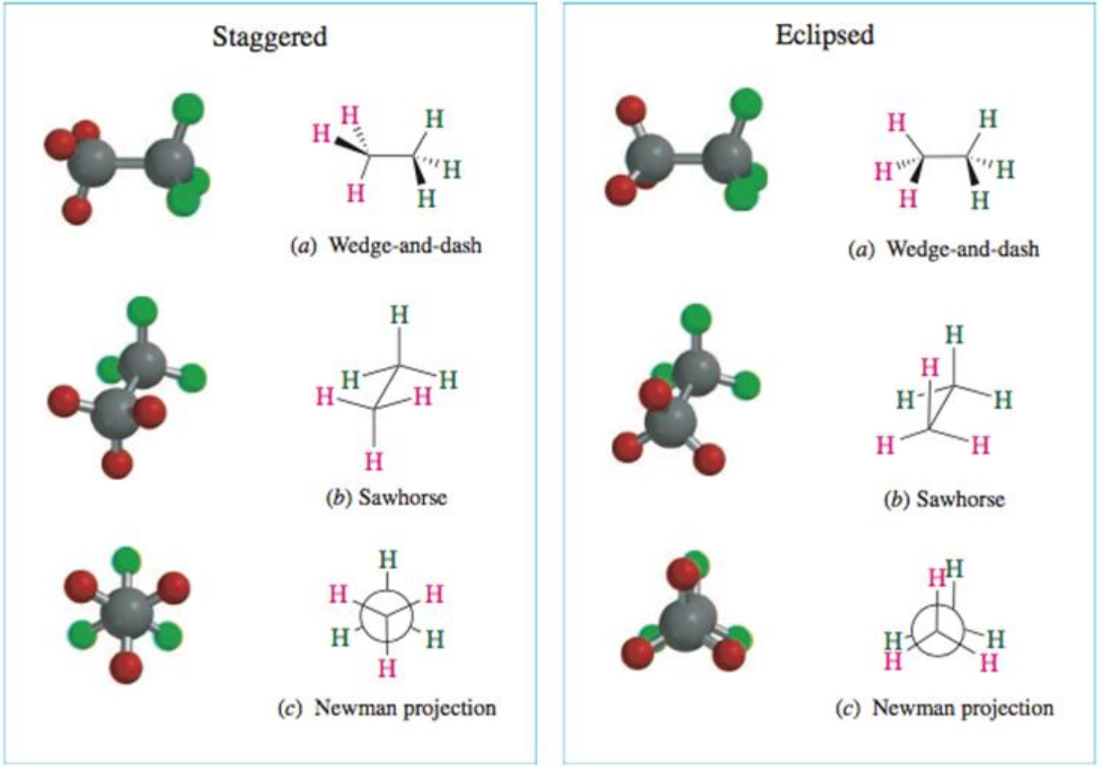
Staggered conformation of ethane



Eclipsed conformation of ethane



Conformations of Ethane-1



Torsion or Dihedral Angles

- We are often interested in the angle between two *bonds* in a conformation (for example, the angle between adjacent C–H bonds in ethane)

- Such an angle is called a **torsion** or **dihedral angle**

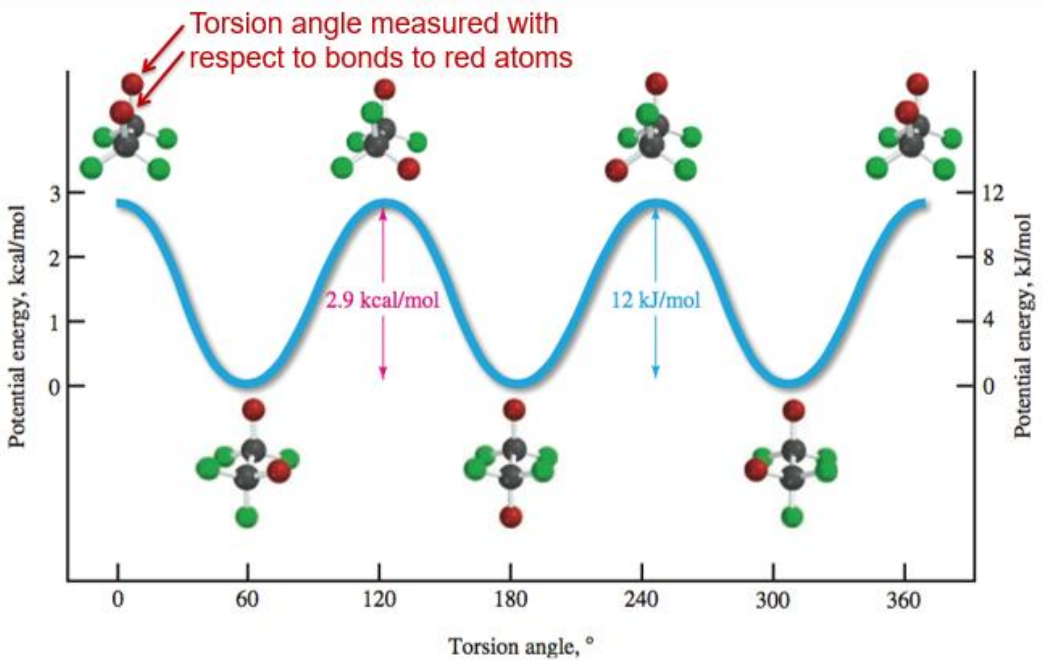
- We depict and visual this by drawing a **Newman Projections**
  - Sight down the C-C bond, front C represented by a dot and bcck C represented by a circle

Certain torsion angles have special names.
Eclipsed = 0°, gauche = 60°, anti = 180°

Newman Projections of Ethane

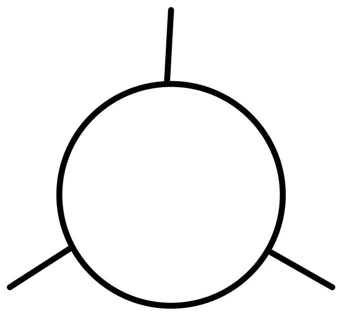
Torsion angle = 0°	Torsion angle = 60°	Torsion angle = 180°
Eclipsed	Gauche	Anti

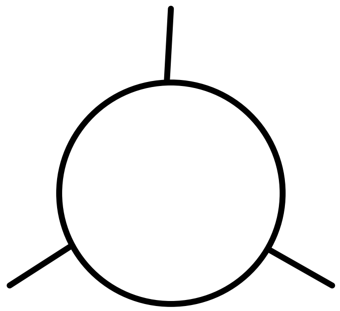
Potential Energy and Conformation

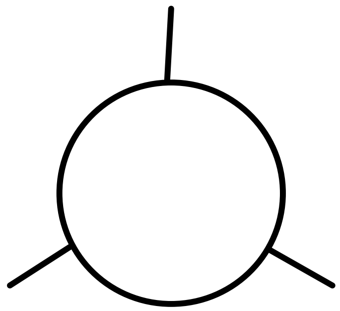


Eclipsed conformers are *least stable*; staggered conformers are *most stable.*

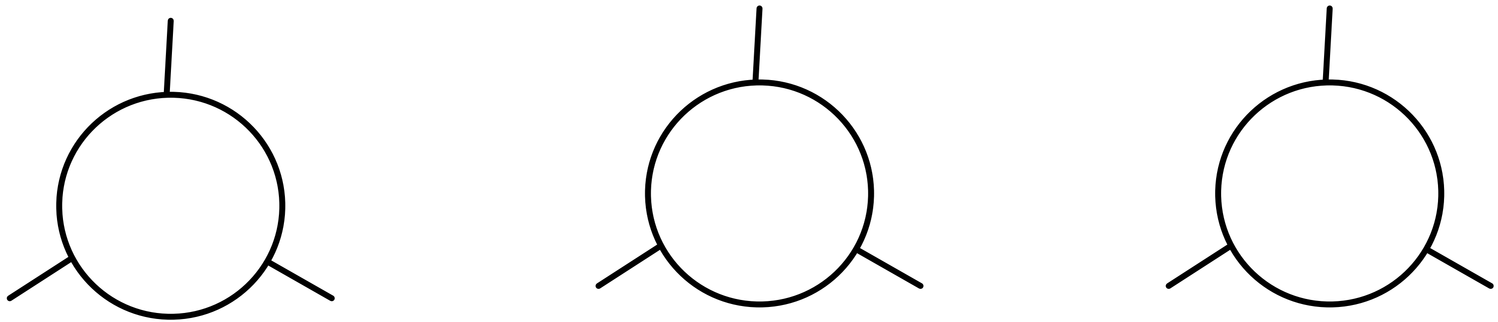
Conformation of Butane



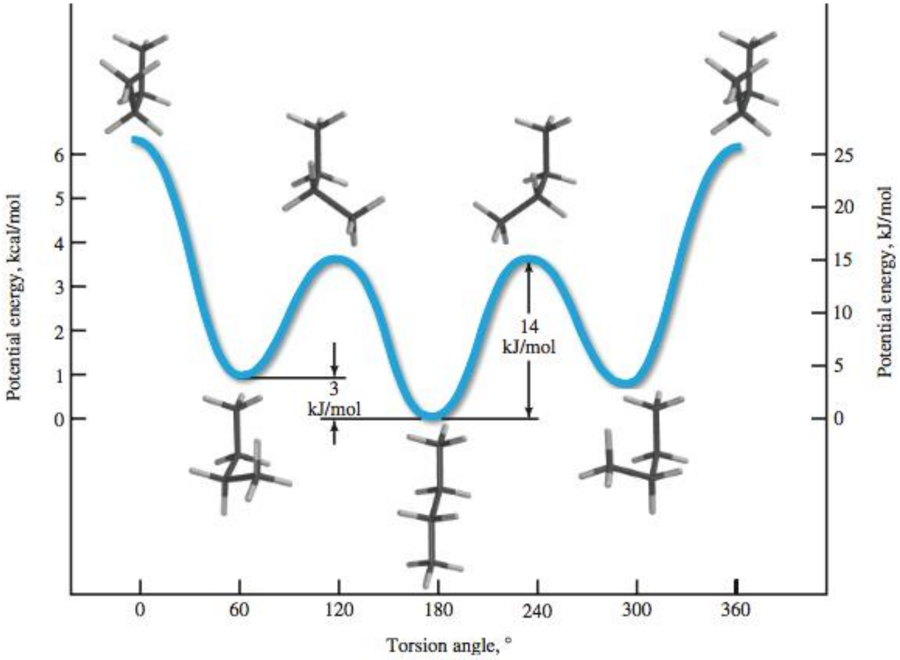




Conformation of Butane-cont.



Potential Energy and Conformation of Butane



Note the impact of the *gauche* interaction on the *gauche* conformer!

Potential Energy Butane

- Staggered conformers are still energy *minima* and eclipsed conformers are still *maxima*

- The *anti* conformer is lower in energy than the *gauche* conformer due to a destabilizing steric interaction in the latter

- Eclipsing CH_3 groups cause greater destabilization than H-H or $H-CH_3$ eclipsing

Alkane Conformation Problem

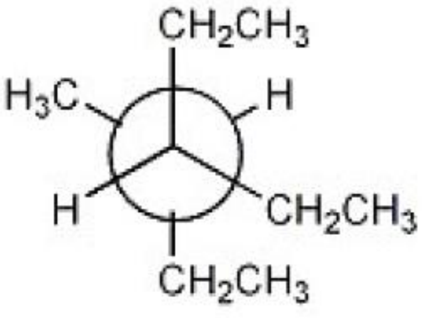
- Draw 2-methylpentane in its most stable conformation sighting down C-2 to C-3.

Alkane Conformation Problem-1

- Draw 2,3-dimethylbutane in its least stable conformation sighting down C-2 to C-3.

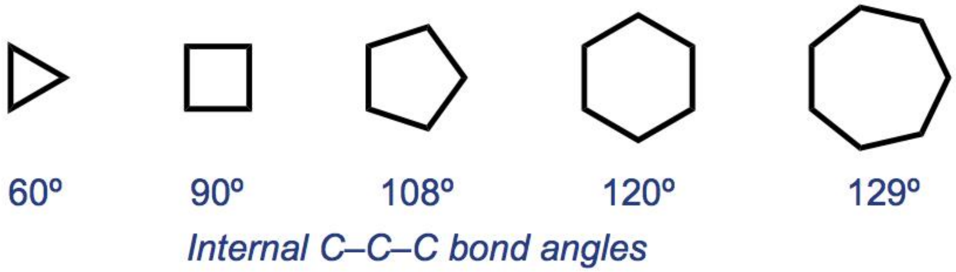
Alkane Conformation Problem-2

- Name the following molecule:



Planar Cycloalkanes: A Prediction

- The most stable cycloalkanes should have bond angles near $109.5°$
- If cycloalkanes are planar, cyclopentane ($108°$) should be most stable, followed by cyclohexane ($120°$)



- We can measure stability using enthalpies of combustion

Heats of Combustion of Cycloalkanes

TABLE 3.1 Heat of combustion ($-\Delta H°$) of Cycloalkanes

Cycloalkane	Number of CH_2 groups	Heat of combustion kJ/mol	Heat of combustion kcal/mol	Heat of combustion per CH_2 group kJ/mol	Heat of combustion per CH_2 group kcal/mol
Cyclopropane	3	2,091	499.8	697	166.6
Cyclobutane	4	2,721	650.3	681	162.7
Cyclopentane	5	3,291	786.6	658	157.3
Cyclohexane	6	3,920	936.8	653	156.0
Cycloheptane	7	4,599	1099.2	657	157.0
Cyclooctane	8	5,267	1258.8	658	157.3

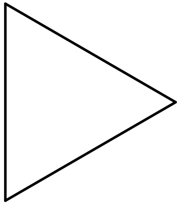
Heats of Combustion of Cycloalkanes

- Cyclopentane is *not* the most stable—cyclohexane is!
- Heat per $\mathrm{CH_2}$ combusted is smallest for cyclohexane

- This is good evidence that cycloalkanes are *not* planar (with the exception of cyclopropane).

Cyclopropane (C_3H_6)

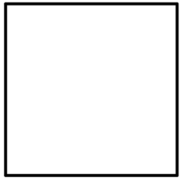
- Cyclopropane is characterized by extremely severe **angle strain**—its internal C–C–C bond angle is only 60°
- The C–C bonds in cyclopropane are *bent* and derived from the angled overlap of sp^3 hybrids
- The internal C–C bonds are weak and liable to break
- Cyclopropane also suffers from **torsional strain** due to eclipsing C–H's

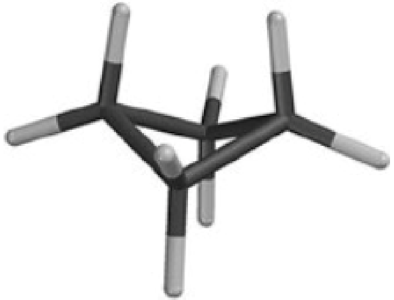


All adjacent pairs
of bonds are eclipsed

Cyclobutane (C_4H_8)

- Angle strain in cyclobutane is not quite as bad as cyclopropane (90° versus 60°)
- In addition, the cyclobutane ring "puckers" to minimize torsional or eclipsing strain

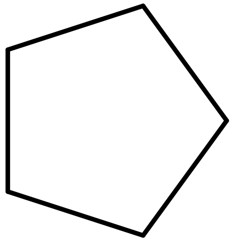


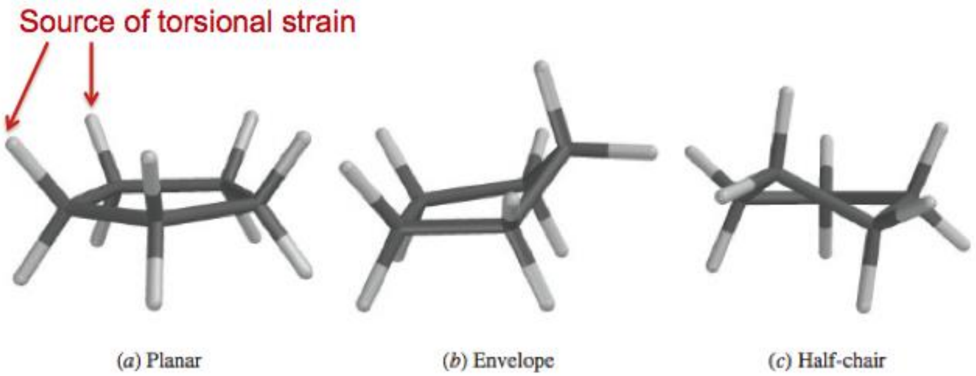


- Nonetheless, cyclobutane is still somewhat unstable

Cyclopentane (C_5H_{10})

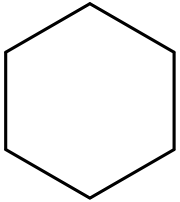
- The planar conformation of cyclopentane has little angle strain, but considerable torsional strain due to eclipsing C–H bonds
- The **envelope** and **half-chair** conformers involve less torsional strain and are of similar energy
- Equilibration between these conformers is rapid

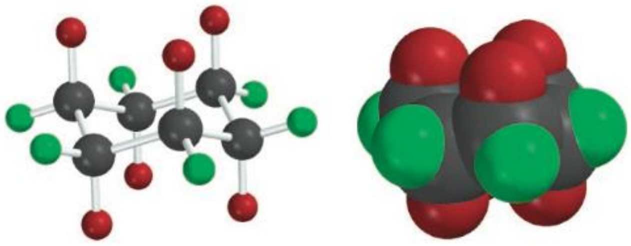




Cyclohexane

- To achieve bond angles of 109.5° and minimize torsional strain, cyclohexane adopts a **chair** conformation

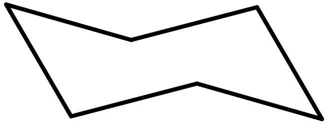


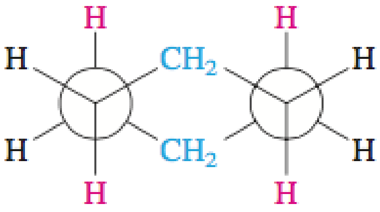


(a) (b)

Chair cyclohexane bears some resemblance to a chaise lounge.

In the chair conformation, all adjacent C–H bonds are staggered

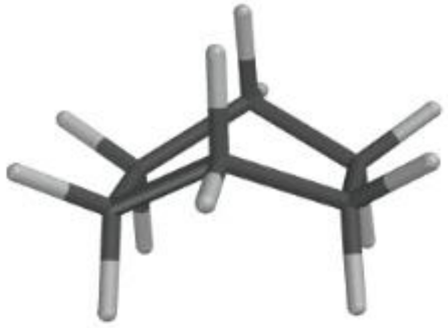




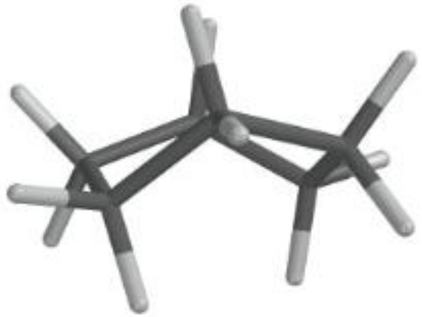
Staggered arrangement of bonds in chair conformation of cyclohexane

Boat and Skew-boat Cyclohexane

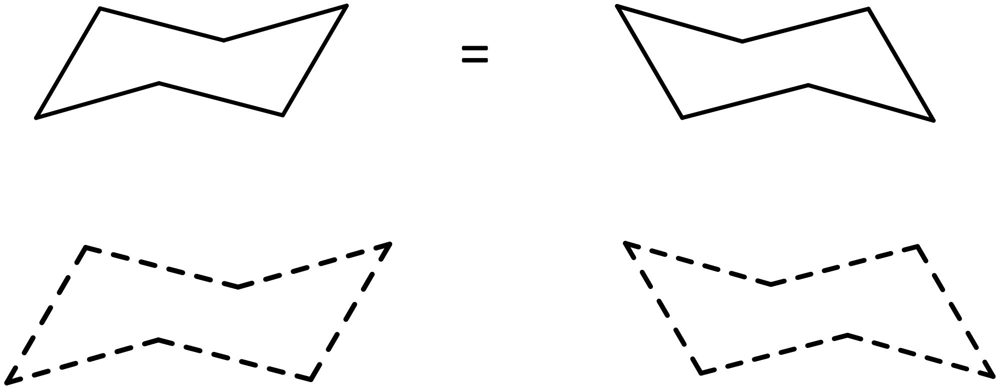
- The **boat** conformer is a high-energy structure containing eclipsing C–H bonds



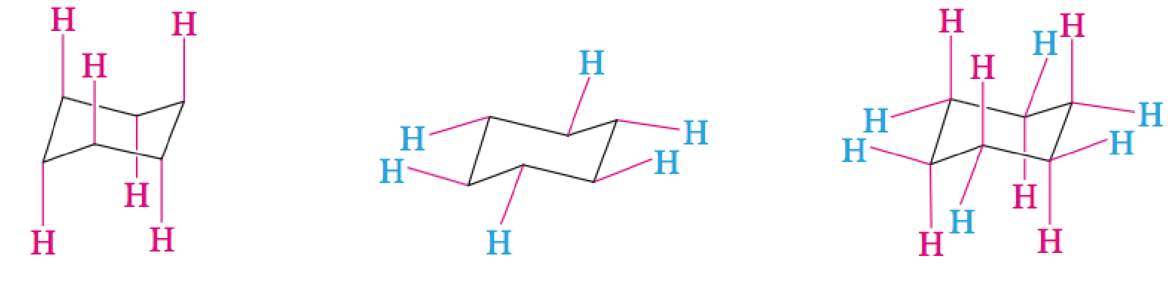
- The **skew-boat** relieves some of the torsional strain in the boat
- Both are considerably higher in energy than the chair conformer



Drawing Chair Conformations

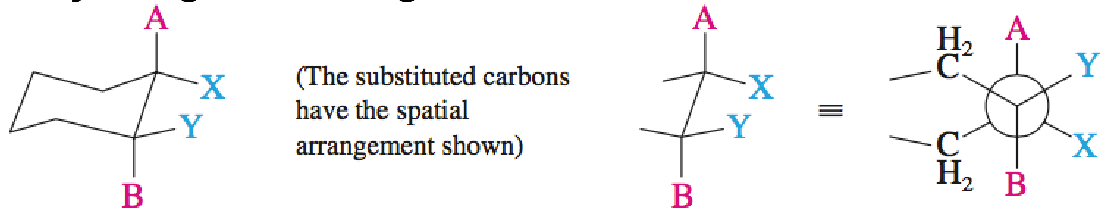


Axial and Equitorial Positions Cyclohexane



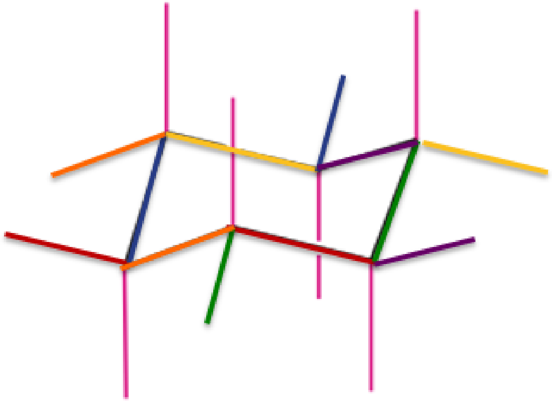
Axial C—H bonds   Equatorial C—H bonds   Axial and equatorial bonds together

- Axial hydrogens are *anti* to one another;
  equatorial hydrogens are *gauche* to one another



Drawing Axial and Equitorial Groups

- Equatorial bonds should be parallel to the ring bonds on the two nearest neighbor carbons

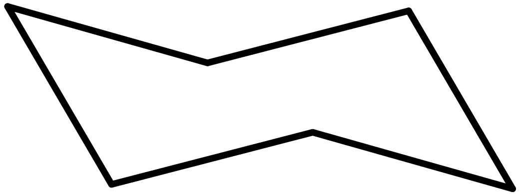


Note how the colored bonds are parallel to one another.

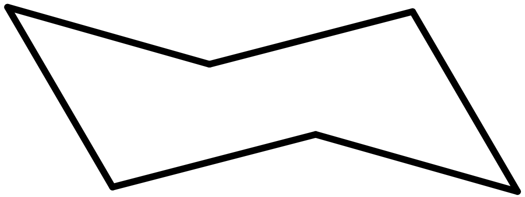
- Drawing the equatorial groups this way ensures that the tetrahedral geometry is represented faithfully

Draw Axial and Equitorial Positions Cyclohexane

Axial Positions

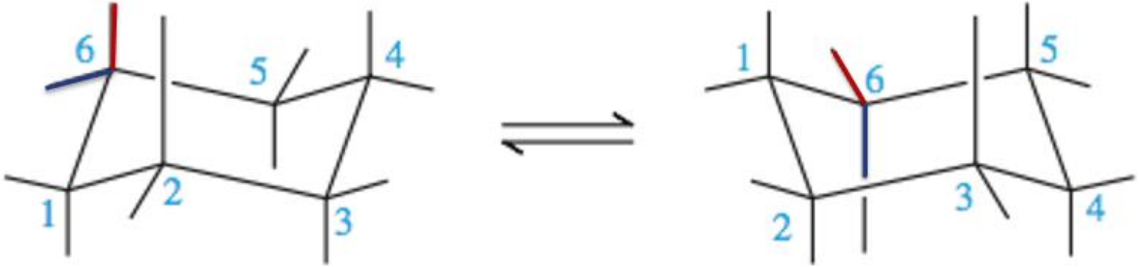


Equatorial Positions



Chair Inversion of Cyclohexane

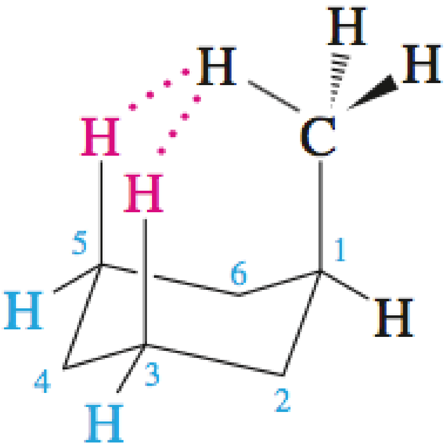
- “Right-leaning” and “left-leaning” chair conformers can interconvert rapidly via **ring inversion** or *chair flip*
- An “up” carbon becomes a “down” carbon and *vice versa* upon inversion
- Axial groups become equatorial and *vice versa*
- However, “up” substituents are still up and “down” substituents are still down!



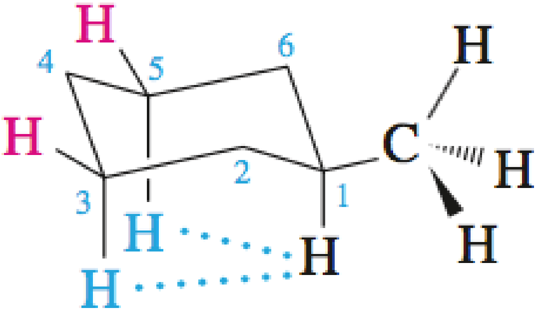
Methylcyclohexane: Most Stable Chair Conformation?

1,3 Diaxial Repulsion

The equatorial conformer lacks destabilizing **1,3-diaxial interactions**.



Van der Waals strain between hydrogen of axial CH_3 and axial hydrogens at C-3 and C-5



Negligible van der Waals strain between hydrogen at C-1 and axial hydrogens at C-3 and C-5

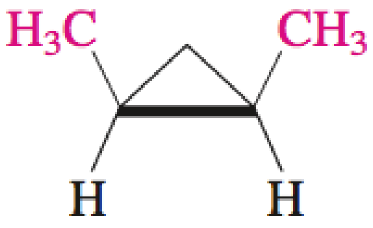
Other Substituted Cyclohexanes

- Other monosubstituted cyclohexanes follow a similar pattern: *the equatorial conformer is more stable than the axial conformer*
- $\Delta G°$ and *K* for interconversion depend on the size of the group (larger group = greater $\Delta G°$ = greater *K*)

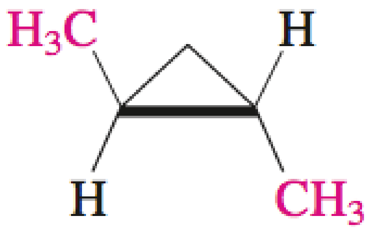
Substituent X	Percent axial	Percent equatorial	*K*
—F	40	60	1.5
—CH_3	5	95	19
—$CH(CH_3)_2$	3	97	32.3
—$C(CH_3)_3$	<0.01	>99.99	>9999

Disubstituted Cycloalkanes

- Disubstitued cycloalkanes containing groups on two different carbons exist as ***cis*** and ***trans*** **isomers**
- The *cis* isomer contains both groups on the same side of the ring
- The *trans* isomer contains the two groups on opposite sides of the ring
- These isomers have the same connectivity but different positions of groups in space (**stereoisomers**)



cis-1,2-Dimethylcyclopropane



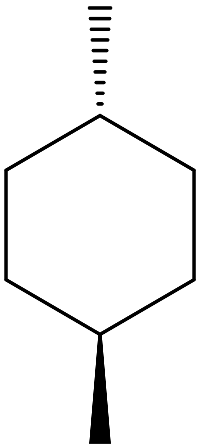
trans-1,2-Dimethylcyclopropane

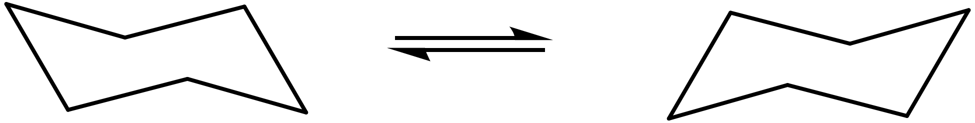
Disubstituted Cyclohexane Conformers

- Substituents in disubstituted cyclohexanes may be axial or equatorial
- To determine which chair conformer of a disubstituted cyclohexane is more stable, we must compare the numbers and sizes of equatorial substituents

Disubstituted Cyclohexane Conformers cont.

- Draw the two chair structures for *trans*-1,4-dimethylcyclohexane. Which one is more stable?





What about cis-1,4-dimethylcyclohexane?

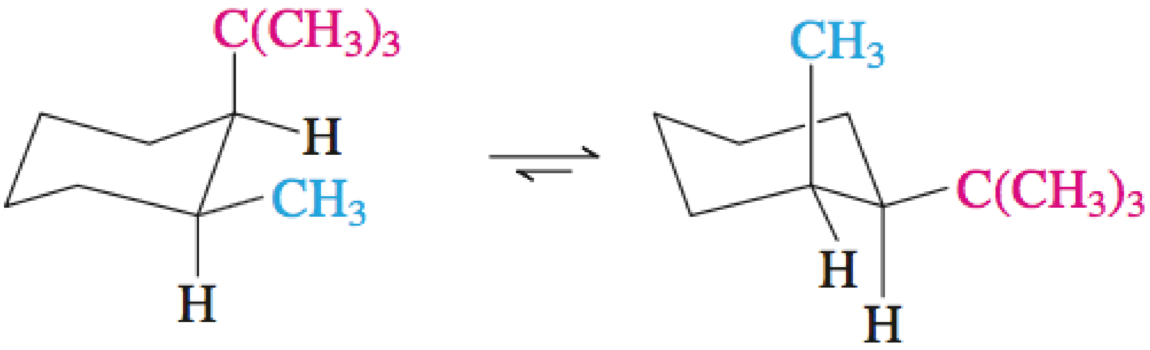
- Draw the most stable conformation isomer.

- Is cis or trans-1,4-dimethylcyclohexane the more stable stereoisomer?

Why does cis-1,3-dichlorocyclohexane have a preferred chair conformation, yet cis-1,4-dichlorocyclohexane does not?

Two Different Substituents

- In disubstituted cyclohexanes with two different substituents, the more stable chair is the one with the larger group equatorial



(Less stable conformation: larger group is axial)

(More stable conformation: larger group is equatorial)

cis-1-*tert*-Butyl-2-methylcyclohexane

Name the compound and draw the most stable chair conformation